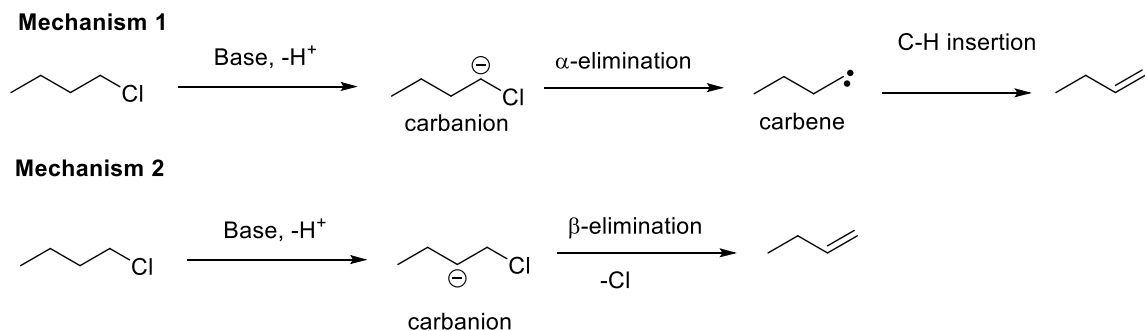


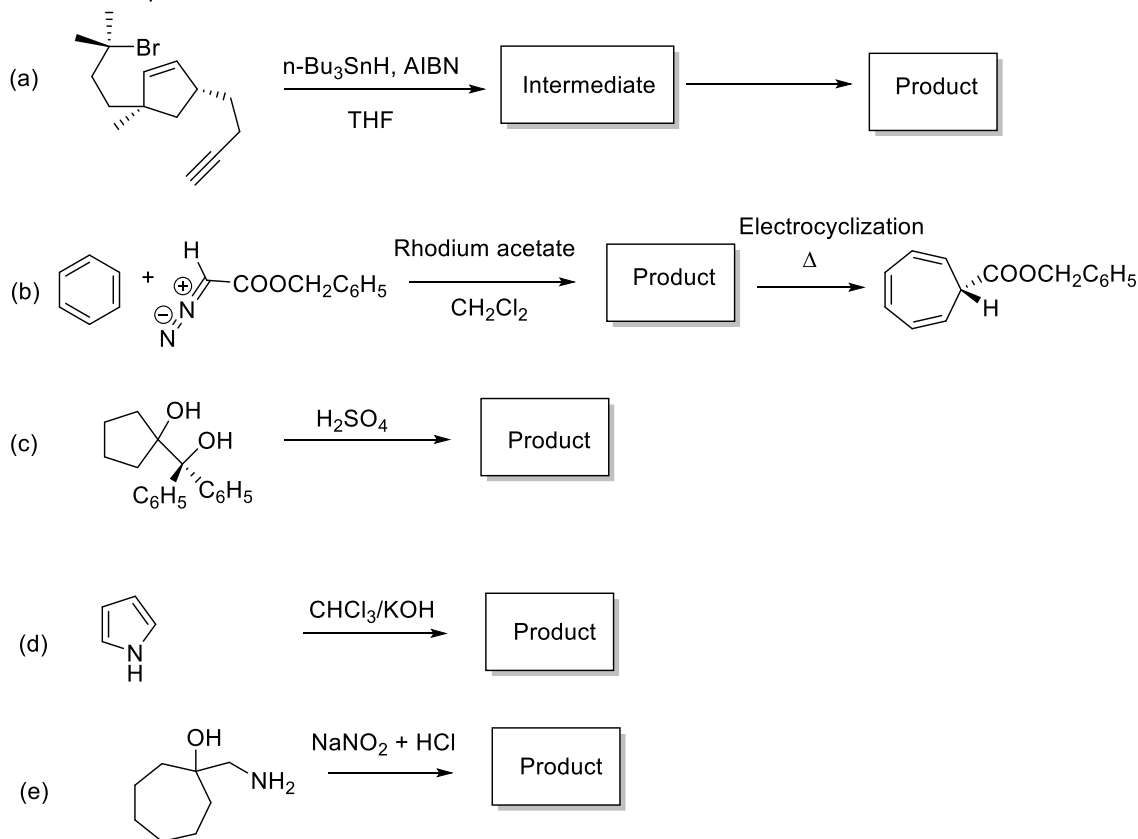
CHM102 Basic Organic Chemistry

Reactive intermediates and reactivity aspects

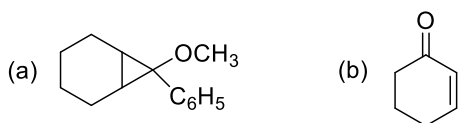
Q1. How to identify the correct mechanism among the following possible pathways?



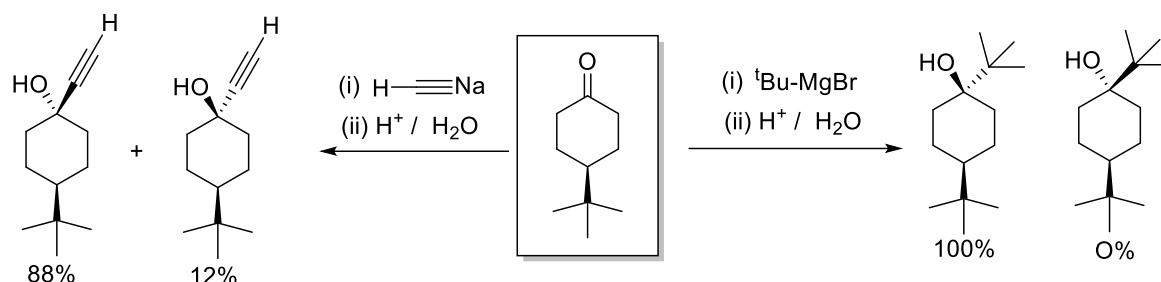
Q2. Identify the product and the reactive intermediate(s) involved in the reactions. Also, indicate arrow-push mechanism for the product formation.



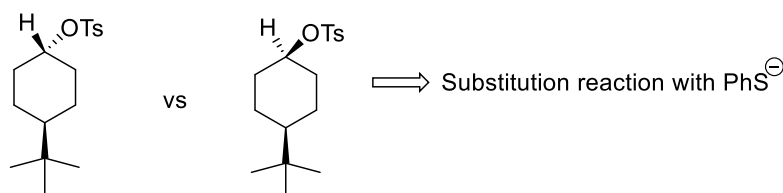
Q3. Provide a synthetic route for accessing the following compounds. Consider any starting material, reactive intermediate(s) and indicate appropriate reaction conditions.



Q4. Explain the reason in nucleophilic addition to the following carbonyl compound show different stereoisomeric products with different nucleophiles? Also, classify those reactions between stereoselective/stereospecific/regioselective/regiospecific reactions.



Q5. Which among the following substrates reacts faster with the given nucleophile? Explain the mechanism with reason.

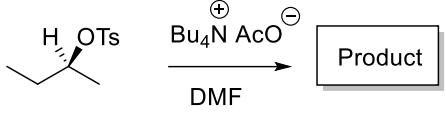
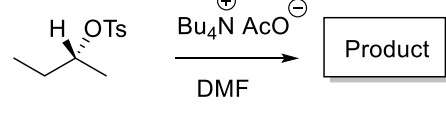
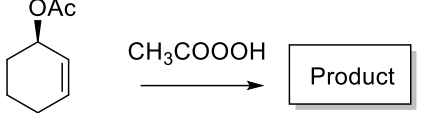
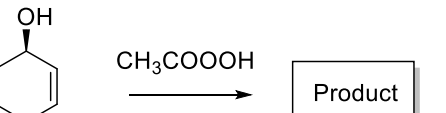
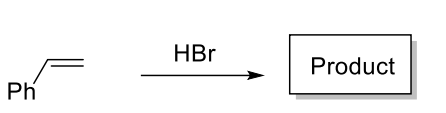
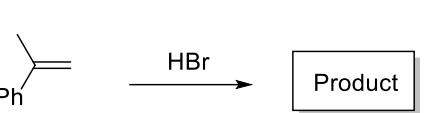
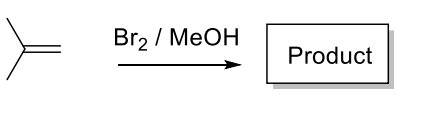
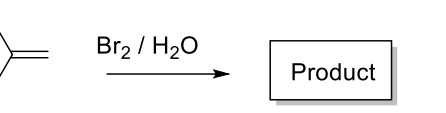
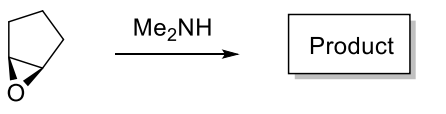
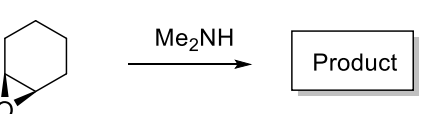
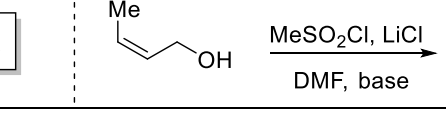
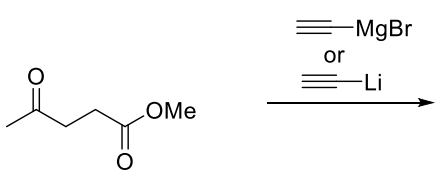
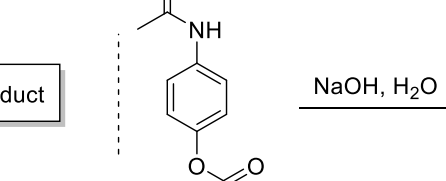
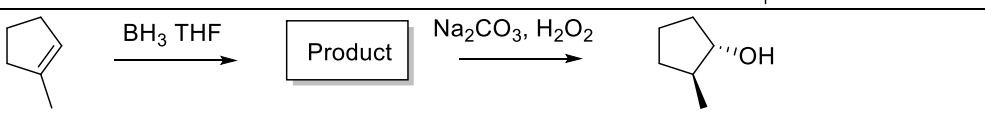


Q6. Predict the major product with reason for each one of the following reactions under the given condition. Also, classify the reaction based on the expected mechanism.

Reactant	Reaction condition	Product	Type of mechanism	Reason
	NaCN			
	NaOCH₃			
	KO-^tBu₃			
	NaI			
	NaOCH₃			
	EtOH			
	CH₃SNa			
	NaOCH₃			
	EtOH			
	NaOCH₃			

Q7. Between *trans* 2-chloro cyclohexanol and *cis* 2-chloro cyclohexanol, one of the isomers only can form cyclohexene oxide under base conditions. Identify the isomer and explain the reason for such behaviour.

Q8. Predict the product and classify the following reaction(s) as stereoselective/stereospecific/regioselective/regiospecific/chemoselectivity with reasons (including arrow-push mechanism).

(a)		
(b)		
(c)		
(d)		
(e)		
(f)		
(g)		
(h)		

Q9. How to synthesize *syn* and *anti* 1,2-diols from a common olefin? Explain the conditions with mechanism.

Q10. Identify the products A and B. Also, explain the reason for the product selectivity in

